Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

Contents

Wednesday 1 April 2026

1.1 Daily Limit

Limit Involving Inverse Hyperbolic and Trigonometric Functions

Let $f(x) = \frac{\tanh^{-1}\left(x\sqrt{1-x^2}\right)}{\tan^{-1}\left(x\sqrt{1-x}\right)}$. Find:

$$\lim_{x\to 0} f(x)$$

Solution: Let $u(x) = x\sqrt{1-x^2}$ and $v(x) = x\sqrt{1-x}$ represent the inner arguments of the inverse functions. Notice that as $x \to 0$, both arguments approach zero:

$$\lim_{x\to 0} u(x) = 0 \quad \text{and} \quad \lim_{x\to 0} v(x) = 0$$

We recall the fundamental limits for the inverse hyperbolic tangent and inverse tangent functions as their arguments approach zero:

$$\lim_{z\to 0} \frac{\tanh^{-1}(z)}{z} = 1 \quad \text{and} \quad \lim_{z\to 0} \frac{\tan^{-1}(z)}{z} = 1$$

We can rewrite the expression for $f(x)$ algebraically by multiplying and dividing by $u(x)$ and $v(x)$:

$$f(x) = \left(\frac{\tanh^{-1}(u(x))}{u(x)}\right) \cdot \left(\frac{v(x)}{\tan^{-1}(v(x))}\right) \cdot \left(\frac{u(x)}{v(x)}\right)$$

Taking the limit as $x \to 0$, we evaluate each of the three factors independently: 1. $\lim_{x\to 0} \frac{\tanh^{-1}(u(x))}{u(x)} = 1$ 2. $\lim_{x\to 0} \frac{v(x)}{\tan^{-1}(v(x))} = 1$ 3. For the algebraic ratio of the arguments:

$$\lim_{x\to 0} \frac{u(x)}{v(x)} = \lim_{x\to 0} \frac{x\sqrt{1-x^2}}{x\sqrt{1-x}} = \lim_{x\to 0} \frac{\sqrt{1-x^2}}{\sqrt{1-x}} = \frac{\sqrt{1-0}}{\sqrt{1-0}} = 1$$

Multiplying the results of these individual limits together yields:

$$L = 1 \cdot 1 \cdot 1 = 1$$

Thursday 2 April 2026

2.1 Daily Limit

Limit of Inverse Tangent with Exponential Argument at Minus Infinity

Evaluate the limit:

$$\lim_{x\to-\infty} 2\frac{\tan^{-1}\left(2^{x-2}\right)}{2^{x-2}}$$

Solution: Let $u = 2^{x-2}$. As $x \to -\infty$, since the base of the exponential function is greater than 1, the exponent approaches $-\infty$, which means:

$$\lim_{x\to-\infty} u = \lim_{x\to-\infty} 2^{x-2} = 0^+$$

We substitute the variable u into our limit expression:

$$L = \lim_{u\to 0^+} 2\frac{\tan^{-1}(u)}{u} = 2 \cdot \lim_{u\to 0^+} \frac{\tan^{-1}(u)}{u}$$

Using the standard trigonometric limit $\lim_{u\to 0} \frac{\tan^{-1}(u)}{u} = 1$, we obtain:

$$L = 2 \cdot 1 = 2$$

Friday 3 April 2026

3.1 Daily Limit

Double Sum Asymptotic Limit via Cotangent Riemann Sum

Evaluate the limit:

$$\lim_{n\to\infty} \sum_{r=2n+1}^{3n} \sum_{k=-6n}^{6n} \frac{1}{r+6kn}$$

Solution: We begin by focusing on the inner summation over the index k. Factoring out $6n$ from the denominator:

$$S_{\text{inner}}(r,n) = \sum_{k=-6n}^{6n} \frac{1}{6n\left(\frac{r}{6n}+k\right)} = \frac{1}{6n} \sum_{k=-6n}^{6n} \frac{1}{\frac{r}{6n}+k}$$

Let $z = \frac{r}{6n}$. Since r ranges from $2n+1$ to $3n$, the variable z lies within the interval $\left(\frac{1}{3}, \frac{1}{2}\right]$. As $n \to \infty$, the symmetric sum over $k \in [-6n, 6n]$ converges uniformly on this interval to the Mittag-Leffler partial fraction expansion of the cotangent function:

$$\lim_{N\to\infty} \sum_{k=-N}^{N} \frac{1}{z+k} = \pi \cot(\pi z)$$

Thus, for large n, the inner sum has the asymptotic form:

$$S_{\text{inner}}(r,n) = \frac{\pi}{6n} \cot\left(\frac{\pi r}{6n}\right) + O\left(\frac{1}{n^2}\right)$$

Substituting this back into the outer summation over r:

$$L = \lim_{n\to\infty} \sum_{r=2n+1}^{3n} \frac{\pi}{6n} \cot\left(\frac{\pi r}{6n}\right)$$

This expression is a standard Riemann sum for a definite integral. Let $t = \frac{r}{n}$, where the grid step is $\Delta t = \frac{1}{n}$. As $n \to \infty$, the discrete summation converts directly into an integral over the interval $[2, 3]$:

$$L = \int_2^3 \frac{\pi}{6} \cot\left(\frac{\pi t}{6}\right)\, dt$$

To evaluate this integral, let $u = \frac{\pi t}{6}$, so $du = \frac{\pi}{6} dt$. The boundaries change as follows:

$$t = 2 \implies u = \frac{\pi}{3}, \qquad t = 3 \implies u = \frac{\pi}{2}$$

Integrating $\cot(u)$:

$$L = \int_{\pi/3}^{\pi/2} \cot(u)\, du = [\ln|\sin(u)|]_{\pi/3}^{\pi/2}$$

Evaluating at the boundaries:

$$L = \ln\left(\sin\frac{\pi}{2}\right) - \ln\left(\sin\frac{\pi}{3}\right) = \ln(1) - \ln\left(\frac{\sqrt{3}}{2}\right) = 0 - \ln\left(\frac{\sqrt{3}}{2}\right) = \ln\left(\frac{2}{\sqrt{3}}\right)$$

Using logarithm properties, this can also be written in the form:

$$L = \frac{1}{2} \ln\left(\frac{4}{3}\right)$$

Saturday 4 April 2026

4.1 Daily Limit

Infinite Series Involving Integrals of Fractional Parts

Evaluate the limit:

$$\lim_{n\to\infty} \sum_{m=0}^{n} \frac{1}{\int_0^{2^m} \{x\}\, dx}$$

where $\{x\} = x - \lfloor x \rfloor$ denotes the fractional part of x.

Solution: First, let us compute the value of the definite integral in the denominator for a fixed non-negative integer m:

$$I_m = \int_0^{2^m} \{x\}\, dx$$

The fractional part function $\{x\}$ is periodic with a fundamental period of 1. Over any unit interval $[k, k+1]$, the integral of $\{x\}$ is identical to the integral of x over $[0, 1]$:

$$\int_0^1 \{x\}\, dx = \int_0^1 x\, dx = \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2}$$

Since the integration domain $[0, 2^m]$ consists of exactly 2^m full unit intervals of length 1, we can split the integral by additivity:

$$I_m = \sum_{k=0}^{2^m-1} \int_k^{k+1} \{x\}\, dx = 2^m \cdot \left(\frac{1}{2}\right) = 2^{m-1}$$

Substituting this result back into the original sum:

$$S_n = \sum_{m=0}^{n} \frac{1}{I_m} = \sum_{m=0}^{n} \frac{1}{2^{m-1}} = \sum_{m=0}^{n} 2^{1-m} = 2\sum_{m=0}^{n} \left(\frac{1}{2}\right)^m$$

Taking the limit as $n \to \infty$, we obtain an infinite geometric series with initial term $a = 2$ and common ratio $r = \frac{1}{2}$:

$$L = \lim_{n\to\infty} S_n = 2\sum_{m=0}^{\infty} \left(\frac{1}{2}\right)^m = 2 \cdot \left(\frac{1}{1 - 1/2}\right) = 2 \cdot 2 = 4$$

Sunday 5 April 2026

5.1 Daily Limit

Fourth Derivative Limit of Sinc-like Inverse Tangent Function

Let $f(x) = \frac{\tan^{-1}(x)}{x}$. Find:

$$\lim_{x\to 0} f^{(4)}(x)$$

Solution: We analyze the function using its Maclaurin series expansion around $x = 0$. Recall the standard Taylor series for the inverse tangent function for $|x| < 1$:

$$\tan^{-1}(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$$

Dividing the entire series term-by-term by x yields the Maclaurin series for $f(x)$:

$$f(x) = \frac{\tan^{-1}(x)}{x} = 1 - \frac{x^2}{3} + \frac{x^4}{5} - \frac{x^6}{7} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n}$$

By defining $f(0) = 1$, the function $f(x)$ becomes analytic (infinitely differentiable) at $x = 0$. By Taylor's theorem, the coefficient of x^k in the Maclaurin expansion of an analytic function is related to its k-th derivative at the origin by:

$$a_k = \frac{f^{(k)}(0)}{k!}$$

In our expansion for $f(x)$, we locate the coefficient of the x^4 term ($k = 4$):

$$a_4 = \frac{1}{5}$$

Equating this coefficient to the derivative formula:

$$\frac{f^{(4)}(0)}{4!} = \frac{1}{5} \implies f^{(4)}(0) = \frac{4!}{5} = \frac{24}{5}$$

Because $f(x)$ is infinitely differentiable at the origin, its fourth derivative $f^{(4)}(x)$ is a continuous function. Therefore, taking the limit as $x \to 0$ simply evaluates to the value at the origin:

$$L = \lim_{x\to 0} f^{(4)}(x) = f^{(4)}(0) = \frac{24}{5}$$

Monday 6 April 2026

6.1 Daily Limit

Right-Hand Derivative Limit of Error Function Expression

Find $\lim_{x\to 0^+} f'(x)$, where:

$$f(x) = \frac{\text{erf}(x)e^{-x}}{\sqrt{1-\sqrt{x}}}$$

Solution: We construct the asymptotic expansion of $f(x)$ for small positive values of x as $x \to 0^+$. We expand each factor in the definition of $f(x)$ independently: 1. The Gauss error function $\text{erf}(x)$ has the Maclaurin series:

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}\,dt = \frac{2}{\sqrt{\pi}}\left(x - \frac{x^3}{3} + O(x^5)\right)$$

2. The exponential factor expands as:

$$e^{-x} = 1 - x + O(x^2)$$

3. The denominator term $(1-x^{1/2})^{-1/2}$ expands using the generalized binomial theorem:

$$(1-x^{1/2})^{-1/2} = 1 + \left(-\frac{1}{2}\right)(-x^{1/2}) + \frac{(-1/2)(-3/2)}{2!}(-x^{1/2})^2 + O(x^{3/2}) = 1 + \frac{1}{2}x^{1/2} + \frac{3}{8}x + O(x^{3/2})$$

Now, multiply these expansions together to obtain the series for $f(x)$:

$$f(x) = \frac{2}{\sqrt{\pi}}\left[x + O(x^3)\right]\cdot\left[1 - x + O(x^2)\right]\cdot\left[1 + \frac{1}{2}x^{1/2} + \frac{3}{8}x + O(x^{3/2})\right]$$

Multiplying through carefully up to order $x^{3/2}$:

$$f(x) = \frac{2}{\sqrt{\pi}}\left(x + \frac{1}{2}x^{3/2} - \frac{5}{8}x^2 + O(x^{5/2})\right)$$

Since this asymptotic series represents an algebraic composition of smooth functions for $x > 0$, we can differentiate term-by-term with respect to x:

$$f'(x) = \frac{2}{\sqrt{\pi}}\left(1 + \frac{3}{4}x^{1/2} - \frac{5}{4}x + O(x^{3/2})\right)$$

Taking the limit of the derivative as $x \to 0^+$ causes all terms containing powers of x to vanish:

$$L = \lim_{x\to 0^+} f'(x) = \frac{2}{\sqrt{\pi}}(1 + 0 - 0 + 0) = \frac{2}{\sqrt{\pi}}$$

Tuesday 7 April 2026

7.1 Daily Limit

Limit of an Exponentiated Product

Evaluate the limit:

$$\lim_{n\to\infty}\left[\prod_{k=1}^{n}\left(\frac{n+k}{n}\right)^{\frac{n^3-2n^2-7n-4}{(n^2-4n)(n^2+2n+1)}}\right]$$

Solution: Notice that the exponent does not depend on the product index k. Therefore, we begin by simplifying the rational expression in the exponent:

$$E(n)=\frac{n^3-2n^2-7n-4}{(n^2-4n)(n^2+2n+1)}$$

We factor both the numerator and the denominator completely: 1. For the numerator, checking roots reveals that $n=-1$ is a root:

$$n^3-2n^2-7n-4=(n+1)(n^2-3n-4)=(n+1)^2(n-4)$$

2. For the denominator, we factor each quadratic directly:

$$(n^2-4n)(n^2+2n+1)=n(n-4)(n+1)^2$$

Substituting these factorizations back into $E(n)$ results in immense algebraic cancellation:

$$E(n)=\frac{(n+1)^2(n-4)}{n(n-4)(n+1)^2}=\frac{1}{n}$$

Our limit expression simplified dramatically:

$$L=\lim_{n\to\infty}\left[\prod_{k=1}^{n}\left(1+\frac{k}{n}\right)\right]^{\frac{1}{n}}$$

To evaluate this limit, we take the natural logarithm of the expression inside the limit, converting the product into a sum:

$$\ln(L)=\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}\ln\left(1+\frac{k}{n}\right)$$

This is a standard Riemann sum with step size $\Delta x=\frac{1}{n}$ and grid points $x_k=\frac{k}{n}$ over the domain $[0,1]$. As $n\to\infty$, it converges to the definite integral:

$$\ln(L)=\int_0^1\ln(1+x)\,dx$$

Using integration by parts (or the standard antiderivative $\int\ln(u)\,du=u\ln(u)-u$):

$$\int_0^1\ln(1+x)\,dx=[(1+x)\ln(1+x)-(1+x)]_0^1$$

Evaluating at the boundaries:

$$\ln(L)=(2\ln(2)-2)-(1\ln(1)-1)=2\ln(2)-1=\ln(4)-\ln(e)=\ln\left(\frac{4}{e}\right)$$

Exponentiating both sides gives the final value of the limit:

$$L=e^{\ln(4/e)}=\frac{4}{e}$$

Wednesday 8 April 2026

8.1 Daily Limit

Trigonometric and Inverse Tangent Limit

Evaluate the limit:

$$\lim_{x\to 0} \sin(x) \frac{\tan^{-1}(x)}{1-\cos(x)}$$

Solution: We analyze the asymptotic behavior of each elementary function in the expression as $x \to 0$. Recall the standard first-order Maclaurin approximations for small angles: 1. For the sine function: $\sin(x) \sim x$ 2. For the inverse tangent function: $\tan^{-1}(x) \sim x$ 3. For the cosine function subtraction: $1 - \cos(x) \sim \frac{x^2}{2}$

We substitute these asymptotic equivalents directly into the limit quotient:

$$L = \lim_{x\to 0} \frac{\sin(x) \cdot \tan^{-1}(x)}{1-\cos(x)} = \lim_{x\to 0} \frac{x \cdot x}{\frac{x^2}{2}}$$

Simplifying the algebraic fraction:

$$L = \lim_{x\to 0} \frac{x^2}{\frac{x^2}{2}} = \lim_{x\to 0} 2 = 2$$

Thursday 9 April 2026

9.1 Daily Limit

High-Order Taylor Cancellation Limit

Evaluate the limit:

$$\lim_{x\to 0} \frac{\sin(\ln(1+x)) - \ln(1+\sin(x))}{x^3}$$

Solution: To determine the behavior of the numerator, we perform a Taylor series expansion of both composite functions around $x = 0$ up to $O(x^4)$. 1. **First term:** $\sin(\ln(1+x))$ We compose the expansions $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + O(x^4)$ and $\sin(u) = u - \frac{u^3}{6} + O(u^5)$:

$$\sin(\ln(1+x)) = \left(x - \frac{x^2}{2} + \frac{x^3}{3}\right) - \frac{1}{6}\left(x - \frac{x^2}{2} + \frac{x^3}{3}\right)^3 + O(x^4)$$

Expanding the cubic term and collecting coefficients up to order x^3:

$$\sin(\ln(1+x)) = x - \frac{x^2}{2} + \left(\frac{1}{3} - \frac{1}{6}\right)x^3 + O(x^4) = x - \frac{x^2}{2} + \frac{x^3}{6} + O(x^4)$$

2. **Second term:** $\ln(1+\sin(x))$ We compose the expansions $\sin(x) = x - \frac{x^3}{6} + O(x^5)$ and $\ln(1+v) = v - \frac{v^2}{2} + \frac{v^3}{3} + O(v^4)$:

$$\ln(1+\sin(x)) = \left(x - \frac{x^3}{6}\right) - \frac{1}{2}\left(x - \frac{x^3}{6}\right)^2 + \frac{1}{3}\left(x - \frac{x^3}{6}\right)^3 + O(x^4)$$

Expanding and collecting coefficients up to order x^3:

$$\ln(1+\sin(x)) = x - \frac{1}{2}x^2 + \left(-\frac{1}{6} + \frac{1}{3}\right)x^3 + O(x^4) = x - \frac{x^2}{2} + \frac{x^3}{6} + O(x^4)$$

Now, subtract the second expansion from the first:

$$\sin(\ln(1+x)) - \ln(1+\sin(x)) = \left(x - \frac{x^2}{2} + \frac{x^3}{6}\right) - \left(x - \frac{x^2}{2} + \frac{x^3}{6}\right) + O(x^4) = O(x^4)$$

Notice that the terms of order x^1, x^2, and x^3 cancel identically! Substituting this $O(x^4)$ remainder into our limit:

$$L = \lim_{x\to 0} \frac{O(x^4)}{x^3} = \lim_{x\to 0} O(x) = 0$$

Friday 10 April 2026

10.1 Daily Limit

Limit Involving Nested Radicals and Logarithm of Cosine

Evaluate the limit:

$$\lim_{x\to 0} \frac{\sin(x^2)\left(\sqrt{1-x\sqrt{1-x\sqrt{1-x\dots}}}\right)}{\ln(\cos(x))}$$

Solution: Let $y(x)$ denote the infinite nested radical in the numerator:

$$y(x) = \sqrt{1-x\sqrt{1-x\sqrt{1-x\dots}}}$$

By the self-similar structure of the infinite nesting, $y(x)$ satisfies the algebraic functional equation:

$$y(x) = \sqrt{1 - x \cdot y(x)}$$

Squaring both sides and rearranging terms yields a quadratic equation for $y(x)$:

$$y^2 + xy - 1 = 0$$

As $x \to 0$, this quadratic equation reduces to $y^2 - 1 = 0$. Since the radical represents a strictly positive real quantity for sufficiently small x, we take the positive root:

$$\lim_{x\to 0} y(x) = 1$$

Next, we determine the asymptotic behavior of the remaining trigonometric and logarithmic factors as $x \to 0$: 1. For the sine factor: $\sin(x^2) \sim x^2$ 2. For the logarithmic factor: We expand $\cos(x) = 1 - \frac{x^2}{2} + O(x^4)$ and use $\ln(1+u) \sim u$:

$$\ln(\cos(x)) = \ln\left(1 - \frac{x^2}{2} + O(x^4)\right) \sim -\frac{x^2}{2}$$

We substitute all three asymptotic components back into the original limit quotient:

$$L = \lim_{x\to 0} \frac{\sin(x^2)\cdot y(x)}{\ln(\cos(x))} = \lim_{x\to 0} \frac{x^2 \cdot 1}{-\frac{x^2}{2}}$$

Canceling the common factor of x^2 gives the final result:

$$L = \frac{1}{-\frac{1}{2}} = -2$$